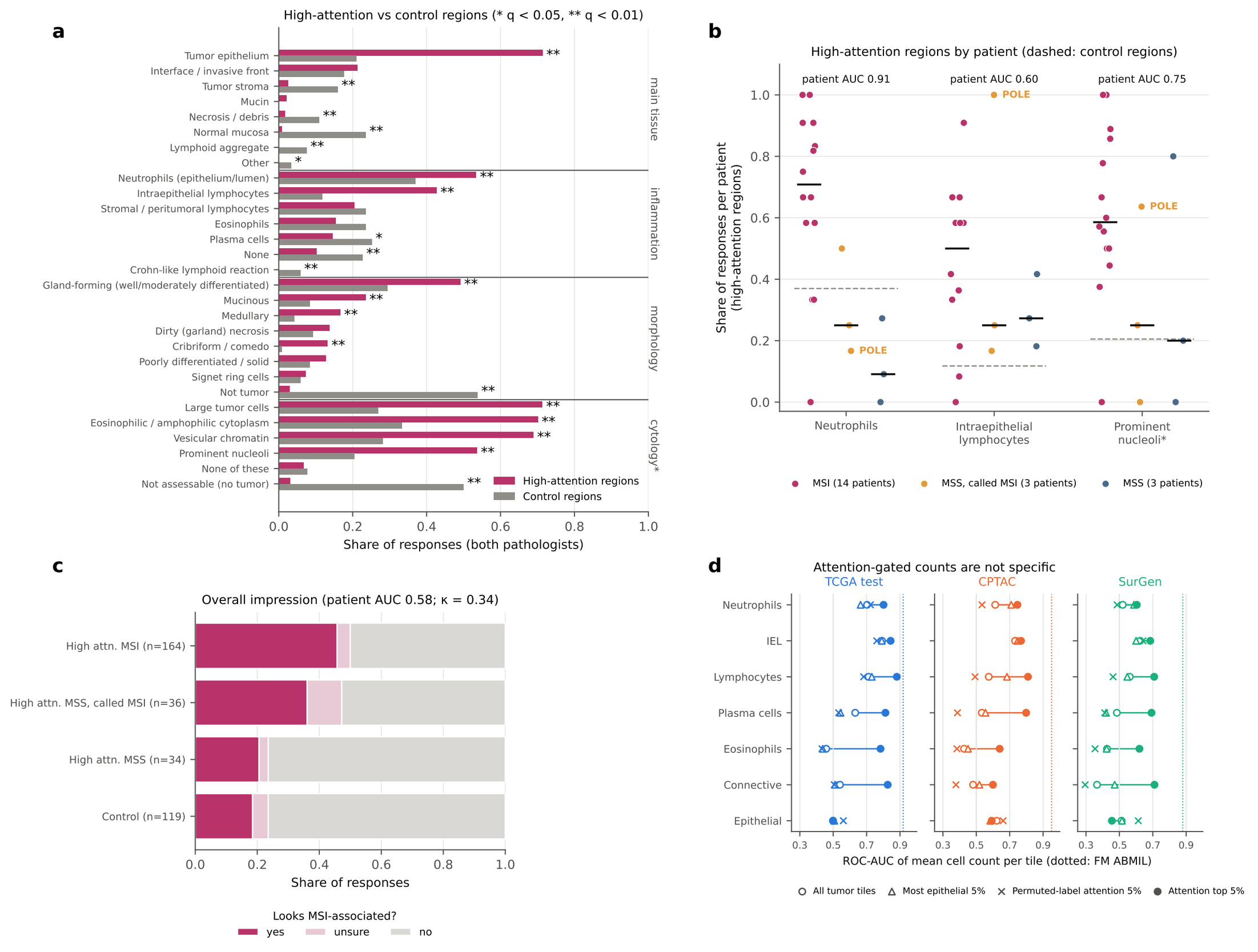
a
High-attention vs control regions (* q < 0.05, ** q < 0.01)
Tumor epithelium
Interface / invasive front
Tumor stroma
Mucin
Necrosis / debris
Normal mucosa
Lymphoid aggregate
Other
Neutrophils (epithelium/lumen)
Intraepithelial lymphocytes
Stromal / peritumoral lymphocytes
Eosinophils
Plasma cells
None
Crohn-like lymphoid reaction
Gland-forming (well/moderately differentiated)
Mucinous
Medullary
Dirty (garland) necrosis
Cribriform / comedo
Poorly differentiated / solid
Signet ring cells
Not tumor
Large tumor cells
Eosinophilic / amphophilic cytoplasm
Vesicular chromatin
Prominent nucleoli
None of these
Not assessable (no tumor)
main tissue
inflammation
morphology
cytology*
High-attention regions
Control regions
0.0
0.2
0.4
0.6
0.8
1.0
Share of responses (both pathologists)
b
High-attention regions by patient (dashed: control regions)
patient AUC 0.91
patient AUC 0.60
patient AUC 0.75
POLE
Share of responses per patient (high-attention regions)
Neutrophils
Intraepithelial lymphocytes
Prominent nucleoli*
MSI (14 patients)
MSS, called MSI (3 patients)
MSS (3 patients)
c
Overall impression (patient AUC 0.58; κ = 0.34)
High attn. MSI (n=164)
High attn. MSS, called MSI (n=36)
High attn. MSS (n=34)
Control (n=119)
Share of responses
Looks MSI-associated?
yes
unsure
no
d
Attention-gated counts are not specific
TCGA test
CPTAC
SurGen
Neutrophils
IEL
Lymphocytes
Plasma cells
Eosinophils
Connective
Epithelial
ROC-AUC of mean cell count per tile (dotted: FM ABMIL)
All tumor tiles
Most epithelial 5%
Permuted-label attention 5%
Attention top 5%